

Kelly's Lemma and Graph Reconstruction

A Worked Example of Defective Calculus

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Claim boundary. This document is a worked example. It proves Kelly’s lemma by a standard double count and then translates the result into Defective Calculus. It does not prove the graph reconstruction conjecture. It does not introduce a new graph invariant. It records how DA describes the projection, fiber, retained kernel, section, and unresolved ambiguity of the deck map.

Abstract

For a finite simple graph G , the vertex-deleted deck is the multiset of cards $G - v$. Kelly’s lemma states that every proper subgraph count is determined by the deck. In defective-calculus terms, the deck map is a θ -projection and the inverse image of a deck is the retained reconstruction fiber. A section of the deck map chooses a representative from that fiber. The graph reconstruction conjecture is exactly the claim that the relevant deck fibers are singletons for graphs with at least three vertices.

1 Standard graph-theoretic setup

All graphs in this document are finite, simple, and undirected. For a graph G , write $V(G)$ for its vertex set and $|G| = |V(G)|$.

Let $\mathcal{U}_n$ be the set of isomorphism classes of n -vertex finite simple graphs. Write $[G] \in \mathcal{U}_n$ for the isomorphism class of G .

Definition 1 (Vertex-deleted deck). *For an n -vertex graph G , its vertex-deleted deck is the multiset*

$$\text{Deck}(G) = \{[G - v] : v \in V(G)\},$$

where multiplicity is counted.

Definition 2 (Deck map). *The deck map is*

$$\delta_n : \mathcal{U}_n \rightarrow \text{MSet}_n(\mathcal{U}_{n-1}), \quad \delta_n([G]) = \text{Deck}(G).$$

When convenient, the codomain is restricted to $\text{im}(\delta_n)$.

Definition 3 (Hypomorphism and deck fiber). *Two n -vertex graphs G and H are hypomorphic when*

$$\text{Deck}(G) = \text{Deck}(H).$$

For a realized deck $D \in \text{im}(\delta_n)$, the deck fiber is

$$\text{Fib}(D) = \delta_n^{-1}(D) = \{[H] \in \mathcal{U}_n : \delta_n([H]) = D\}.$$

Definition 4 (Reconstructible parameter). *A graph parameter $p : \mathcal{U}_n \rightarrow A$ is reconstructible from the deck if there exists a map*

$$\tilde{p} : \text{im}(\delta_n) \rightarrow A$$

such that

$$p = \tilde{p} \circ \delta_n.$$

Equivalently, p is constant on every deck fiber.

2 Kelly's lemma

Let $N(F, G)$ denote the number of subgraphs of G isomorphic to F . Let $N_{\text{ind}}(F, G)$ denote the number of induced subgraphs of G isomorphic to F .

Lemma 1 (Kelly's lemma). *Let G be an n -vertex graph and let F be a graph with $m < n$ vertices. Then*

$$N(F, G) = \frac{1}{n - m} \sum_{C \in \text{Deck}(G)} N(F, C),$$

where cards are summed with multiplicity. The same formula holds for induced counts:

$$N_{\text{ind}}(F, G) = \frac{1}{n - m} \sum_{C \in \text{Deck}(G)} N_{\text{ind}}(F, C).$$

Proof. Consider the incidence set

$$\Omega = \{(v, X) : v \in V(G), X \subseteq G, X \cong F, v \notin V(X)\}.$$

Count Ω in two ways.

First fix a copy $X \cong F$ in G . It uses m vertices, so there are $n - m$ vertices outside X . Deleting any outside vertex leaves X visible in the card. Hence

$$|\Omega| = (n - m)N(F, G).$$

Second fix a vertex v . The copies $X \cong F$ avoiding v are exactly the copies of F in $G - v$. Hence

$$|\Omega| = \sum_{v \in V(G)} N(F, G - v) = \sum_{C \in \text{Deck}(G)} N(F, C).$$

Equating the two counts gives the formula. The induced version is identical: an induced copy on a vertex set S survives exactly in the cards obtained by deleting vertices outside S . $\square$

Corollary 1 (Kelly counts factor through the deck). *For every graph F with $|F| < n$, the parameters*

$$[G] \mapsto N(F, G), \quad [G] \mapsto N_{\text{ind}}(F, G)$$

are reconstructible from the deck.

Proof. The displayed formulas express both counts entirely as functions of the multiset of cards. $\square$

3 The deck as a complete $(n - 1)$ -count vector

For $F \in \mathcal{U}_{n-1}$, the multiplicity of F in $\text{Deck}(G)$ is exactly

$$N_{\text{ind}}(F, G).$$

Thus the deck is equivalently the vector of all induced $(n - 1)$ -vertex subgraph counts:

$$\delta_n([G]) = (N_{\text{ind}}(F, G))_{F \in \mathcal{U}_{n-1}}.$$

Kelly's lemma says that this same deck determines every smaller proper subgraph count.

The boundary of the lemma is explicit: if $|F| = n$, the denominator $n - |F|$ vanishes. Kelly's lemma reconstructs proper subgraph counts, not the full graph as a spanning object.

4 Reconstruction as a fiber problem

Definition 5 (Reconstructible graph). *An n -vertex graph G is reconstructible when*

$$\delta_n^{-1}(\text{Deck}(G)) = \{[G]\}.$$

Definition 6 (Graph reconstruction conjecture). *The graph reconstruction conjecture asserts that every finite simple graph with at least three vertices is reconstructible. Equivalently, δ_n is injective for every $n \geq 3$.*

Definition 7 (Section of the deck map). *A section of the deck map is a function*

$$s : \text{im}(\delta_n) \rightarrow \mathcal{U}_n$$

with

$$\delta_n \circ s = \text{id}_{\text{im}(\delta_n)}.$$

Such a section selects one representative isomorphism class from each nonempty deck fiber.

Proposition 1 (Order asymmetry). *For every section s ,*

$$\delta_n \circ s = \text{id}_{\text{im}(\delta_n)}.$$

The opposite composite

$$P_s = s \circ \delta_n$$

is idempotent:

$$P_s^2 = P_s.$$

It equals the identity on $\mathcal{U}_n$ exactly when δ_n is injective.

Proof. The first identity is the definition of a section. Then

$$P_s^2 = (s \circ \delta_n) \circ (s \circ \delta_n) = s \circ (\delta_n \circ s) \circ \delta_n = s \circ \text{id} \circ \delta_n = P_s.$$

If $s \circ \delta_n = \text{id}_{\mathcal{U}_n}$ and $\delta_n([G]) = \delta_n([H])$, then

$$[G] = s(\delta_n([G])) = s(\delta_n([H])) = [H],$$

so δ_n is injective. Conversely, if δ_n is injective, every fiber is a singleton, so every section chooses the unique element in each fiber. $\square$

5 Defective-calculus translation

The graph-theoretic data translate directly into Defective Calculus.

Defective-calculus object	Graph reconstruction object
Carrier $\mathcal{X}$	$\mathcal{U}_n$, isomorphism classes of n -vertex graphs
Equivalence $\sim$	Graph isomorphism
θ -projection D_θ	Deck map δ_n
Observable $D_\theta(G)$	Vertex-deleted deck $\text{Deck}(G)$

Fiber $\mathcal{F}_\theta(D)$	Deck fiber $\delta_n^{-1}(D)$
Retained kernel K_{ret}	Admissible class of representatives inside the deck fiber
α -reconstruction I_α	Section or representative selection from a deck fiber
Polarity σ	Determined when fiber is singleton; unresolved when not
Residual	Failure of selected representative to be forced by the deck

Thus

$$D_\theta([G]) = \delta_n([G]) = \text{Deck}(G).$$

The retained reconstruction class is

$$K_{\text{ret}}([G]) = \delta_n^{-1}(\text{Deck}(G)).$$

When this class is a singleton, the graph is reconstructible. When it is not known to be a singleton, DA records unresolved polarity:

$$\sigma(\text{Deck}(G)) = \perp.$$

6 What Kelly's lemma says in DA language

For every proper graph F with $|F| < n$, the count observable

$$O_F([G]) = N(F, G)$$

factors through the θ -projection:

$$O_F = \tilde{O}_F \circ D_\theta.$$

Therefore all graphs in the same deck fiber have the same proper subgraph counts:

$$D_\theta(G) = D_\theta(H) \implies O_F(G) = O_F(H) \quad \text{for all } |F| < n.$$

In defective-calculus language, proper subgraph counts do not belong to the unresolved kernel. They have already survived the projection. The unresolved kernel, if any, is the residual representative choice inside the deck fiber.

7 Failure surfaces

Code	Failure surface
G1	Treating Kelly's lemma as a proof of graph reconstruction.
G2	Confusing reconstructible parameters with reconstruction of the whole graph.
G3	Introducing labels, roots, or overlap identifications not present in the deck.
G4	Treating a section choice as canonical when the fiber is not known to be a singleton.
G5	Ignoring the boundary $ F = n$, where Kelly's denominator vanishes.

G6

Using non-deck data to collapse $\sigma = \perp$.

8 Known unknowns and proof obligations

1. Determine whether each deck fiber is a singleton for $n \geq 3$.
2. Characterize what additional data specify a representative when a fiber is not known to be a singleton.
3. Compare deck fibers with other graph-observable fibers without conflating their data sources.
4. State every enrichment with a no-leakage rule: if it is not computed from the deck, it is extra data.

9 Final compression

Kelly's lemma states that every proper subgraph count factors through the deck map:

$$N(F, G) = \frac{1}{n - |F|} \sum_{C \in \text{Deck}(G)} N(F, C), \quad |F| < n.$$

The reconstruction problem is the fiber problem

$$\delta_n^{-1}(\text{Deck}(G)) \stackrel{?}{=} \{[G]\}.$$

In DA notation:

$$D_\theta = \delta_n, \quad K_{\text{ret}}(G) = D_\theta^{-1}(D_\theta G), \quad I_\alpha = \text{admissible representative selection.}$$

The worked example shows that DA faithfully locates the retained kernel of graph reconstruction. It does not prove the reconstruction conjecture.

References

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- [3] J. A. Bondy and R. L. Hemminger, Graph reconstruction—a survey, *Journal of Graph Theory* 1 (1977), 227–268.